Menu Driven Program on

Digits of a Number

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Batch	**16**
Date	**7 September 2026**

1. Aim / Objective

To write a menu-driven C program using switch-case that accepts a positive integer from the user and, based on the chosen option, either (1) checks whether all digits of the number are unique, or (2) finds the frequency of occurrence of each digit from 0 to 9 in the number.

2. Algorithm

1. Start.
2. Display the menu with two options:
    i. Check whether all digits are unique
    ii. Find frequency of each digit (0–9)
3. Read the user's menu choice into the variable choice.
4. If choice is neither 1 nor 2, display "Invalid choice!" and go to Step 9.
5. Read a positive integer into the variable number.
6. If number ≤ 0, display "Invalid input!" and go to Step 9.
7. If choice = 1 (Uniqueness Check):
    a. Initialise an array digitSeen[0..9] with all zeros and set isUnique = 1.
    b. Repeat while number ≠ 0: extract digit = number % 10; if digitSeen[digit] is already 1, set isUnique = 0 and stop the loop; otherwise set digitSeen[digit] = 1 and update number = number / 10.
    c. If isUnique = 1, print "Unique"; otherwise print "Not Unique".
8. If choice = 2 (Frequency Count):
    a. Initialise an array freq[0..9] with all zeros.
    b. Repeat while number ≠ 0: extract digit = number % 10; increment freq[digit]; update number = number / 10.
    c. Print freq[i] for i = 0 to 9.
9. Stop.

3. Test Case Table

The program was compiled and executed for the following test cases, covering both valid and invalid inputs:

Test No.	Menu Choice	Input Number	Expected Output	Actual Output	Result
1	1 (Unique Check)	`12345`	`Unique`	`Unique`	Pass
2	1 (Unique Check)	`12234`	`Not Unique`	`Not Unique`	Pass

Test No.	Menu Choice	Input Number	Expected Output	Actual Output	Result
3	2 (Frequency)	`112333`	`freq_0=0, freq_1=2, freq_2=1, freq_3=3, freq_4=0 ... freq_9=0`	`freq_0=0, freq_1=2, freq_2=1, freq_3=3, freq_4=0 ... freq_9=0`	**Pass**
4	5 (Invalid)	`100`	`Invalid choice! Please enter 1 or 2.`	`Invalid choice! Please enter 1 or 2.`	**Pass**
5	1 (Unique Check)	`-50`	`Invalid input! Please enter a positive integer.`	`Invalid input! Please enter a positive integer.`	**Pass**

4. Source Code (C Program)

```c
/* ============================================================
   Program    : Menu Driven Program on Digits of a Number
   Description: Accepts a positive integer from the user and
                performs one of the following operations chosen
                through a switch-case menu:
                1. Check whether all digits of the number are
                   unique.
                2. Find the frequency of each digit (0-9) in
                   the number.
   ============================================================ */

#include <stdio.h>

int main()
{
    int choice;
    long long number;

    printf("=====================================\n");
    printf("   MENU DRIVEN PROGRAM ON DIGITS\n");
    printf("=====================================\n");
    printf("1. Check if all digits are unique\n");
    printf("2. Find frequency of each digit (0-9)\n");
    printf("=====================================\n");
    printf("Enter your choice (1/2): ");
    scanf("%d", &choice);

    /* Input validation for menu choice */
    if (choice != 1 && choice != 2)
    {
        printf("Invalid choice! Please enter 1 or 2.\n");
        return 0;
    }

    printf("Enter a positive integer: ");
    scanf("%lld", &number);

    /* Input validation for positive integer */
    if (number <= 0)
    {
        printf("Invalid input! Please enter a positive integer.\n");
        return 0;
    }

    switch (choice)
    {
        case 1:
        {
            /* Check whether all digits are unique */
            int digitSeen[10] = {0};   /* to mark digits already found */
            int isUnique = 1;          /* flag: 1 = unique so far */
            long long temp = number;
            int digit;

            while (temp != 0)
```

```
            {
                digit = temp % 10;
                if (digitSeen[digit] == 1)
                {
                    isUnique = 0;      /* digit repeated */
                    break;
                }
                digitSeen[digit] = 1;
                temp = temp / 10;
            }

            printf("\nNumber: %lld\n", number);
            if (isUnique == 1)
                printf("Output: Unique\n");
            else
                printf("Output: Not Unique\n");

            break;
        }

        case 2:
        {
            /* Find frequency of each digit 0-9 */
            int freq[10] = {0};
            long long temp = number;
            int digit, i;

            while (temp != 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            printf("\nNumber: %lld\n", number);
            for (i = 0; i <= 9; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }

            break;
        }
    }

    return 0;
}
```

5. Compilation & Execution (Terminal Output)

The program was compiled using GCC and executed in the VS Code integrated terminal (Windows PowerShell) for each test case:

```
PS C:\Users\Rudra Pratap Singh> gcc digit_ops.c -o digit_ops.exe
PS C:\Users\Rudra Pratap Singh> ./digit_ops.exe
=====================================
   MENU DRIVEN PROGRAM ON DIGITS
=====================================
1. Check if all digits are unique
2. Find frequency of each digit (0-9)
=====================================
Enter your choice (1/2): 1
Enter a positive integer: 12345

Number: 12345
Output: Unique

PS C:\Users\Rudra Pratap Singh> ./digit_ops.exe
=====================================
   MENU DRIVEN PROGRAM ON DIGITS
=====================================
1. Check if all digits are unique
2. Find frequency of each digit (0-9)
=====================================
Enter your choice (1/2): 1
Enter a positive integer: 12234

Number: 12234
Output: Not Unique

PS C:\Users\Rudra Pratap Singh> ./digit_ops.exe
=====================================
   MENU DRIVEN PROGRAM ON DIGITS
=====================================
1. Check if all digits are unique
2. Find frequency of each digit (0-9)
=====================================
Enter your choice (1/2): 2
Enter a positive integer: 112333

Number: 112333
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
```

```
freq_9 = 0

PS C:\Users\Rudra Pratap Singh> ./digit_ops.exe
=====================================
   MENU DRIVEN PROGRAM ON DIGITS
=====================================
1. Check if all digits are unique
2. Find frequency of each digit (0-9)
=====================================
Enter your choice (1/2): 5
Invalid choice! Please enter 1 or 2.

PS C:\Users\Rudra Pratap Singh> ./digit_ops.exe
=====================================
   MENU DRIVEN PROGRAM ON DIGITS
=====================================
1. Check if all digits are unique
2. Find frequency of each digit (0-9)
=====================================
Enter your choice (1/2): 1
Enter a positive integer: -50
Invalid input! Please enter a positive integer.
PS C:\Users\Rudra Pratap Singh>
```

6. Result / Conclusion

The menu-driven C program was successfully written, compiled, and executed using switch-case statements. Both operations — checking digit uniqueness and computing digit frequency — were verified against the prepared test cases (including invalid inputs), and the actual output matched the expected output in every case. The program is therefore working correctly.